Rotaract Club of Xiamen University Malaysia

Rotary Club of Kuala Lumpur Dynamics

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SDC Hackathon

Phase 1 : Preliminary Project Proposal

Hosted by Xiamen University Malaysia (XMUM) AI Club

Submission Deadline: 25 October 2026

Instructions: Complete every field below and submit as a single PDF document named “SDC2026_TeamName_Preliminary.pdf” before the deadline above. Answers must comply with the word limits stated for each open-ended question. Incomplete submissions may not be eligible for review.

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1. Team & Project Overview

Team Name	Prompt Masters
Selected Domain	Domain 2
Selected SDG Track (1–17)	SDG 8 - Decent work and economic growth
Project / Solution Name	EntreMaster
Name (Leader)	Lian Yi Yen
Email & Phone (Leader)	yiyenlian7@gmail.com & 01156209436
University Affiliation(s)	University Malaya
Number of Team Members	5

2. Executive Summary & Problem Statement

2.1 The Problem Context

Describe the specific, real-world challenge your team is tackling. Who is affected by this problem, and why does it matter? Highlight the critical social, environmental, or economic issues involved.

[Maximum 250 words]

Small and medium enterprises (SMEs) and localized retailers in Malaysia face severe operational inefficiencies in inventory management due to reliance on manual spreadsheets or outdated, insecure desktop software. Over-stocking causes financial capital lockup and significant material/resource waste, while under-stocking leads to stockouts, customer dissatisfaction, and revenue loss. Furthermore, existing cloud platforms are often unaffordable for small businesses and lack strict multi-tenant security boundaries, exposing operational data to cyber threats or unauthorized cross-tenant data leaks. According to regional supply chain reports, poorly optimized SMB inventory processes result in up to 20–30% resource waste annually. EntreMaster addresses this challenge by providing an affordable, highly secure, AI-ready multi-tenant inventory management system that eliminates stock waste, optimizes reorder thresholds, and strictly enforces enterprise-grade security for small businesses.

2.2 The Proposed AI Solution

Provide a high-level overview of your proposed application. How does it apply artificial intelligence to solve the problem stated above? Briefly mention the tools or technical skills gained from the SDC Phase 1 workshops (e.g., GenAI APIs, database architecture, cloud hosting) that you intend to leverage.

[Maximum 300 words]

EntreMaster is a modern web-based inventory management platform built for multi-tenant SME deployment. It combines real-time inventory tracking, multi-tenant role-based access control (RBAC), and automated reorder alerts with a security-first architecture. By applying smart analytical algorithms and GenAI assistance, EntreMaster automatically analyzes inventory turnover rates, detects stocking anomalies, and assists warehouse managers in generating optimal reorder schedules to prevent overstocking and waste.

3. Technical Architecture & Implementation Plan

3.1 Conceptual Tech Stack

List the technologies, frameworks, APIs, and databases you plan to use to build a functional, deployed prototype. Describe (or attach as an appendix) a simple system block diagram showing how the workshop elements connect into one architecture.

Did your project utilize any official SDC sponsor-provided GenAI API or Cloud Platform? If yes, specify which one and how it is used.

- Technologies & Frameworks: Node.js (Express.js), HTML5/JavaScript Frontend, PostgreSQL 17 (Supabase Platform).
- Security & Auth: JSON Web Tokens (jsonwebtoken), Row-Level Security (RLS) policies, SSL/TLS database transport.
- DevOps & Testing: Replit Secrets Manager, Pytest/Node offline automated security suite (run-phase2-isolated.py, security.test.cjs).
- System Block Diagram Description:

  - Frontend (Web UI): User requests → JWT Authentication Header.

  - Backend (Node.js/Express): Request validation → Auth & Authorization middleware (authorize.cjs, verify-token.cjs) → Pool management (pools.cjs).

  - Database (PostgreSQL / Supabase): RLS policy checks per tenant ID → Safe CRUD transactions → SSL-encrypted response.

- Sponsor-Provided Resources: None (Built entirely using standard open-source tools and free-tier infrastructure).

3.2 Design-Thinking & Deployment Strategy

Explain how your team applied design-thinking methodology to ensure this solution is technically viable, user-friendly, and ready for real deployment by Phase 2.

[Maximum 200 words]

We applied Design-Thinking by conducting user journey mapping for SMB warehouse managers. Empathize & Define: Identified core pain points—complex UI onboarding, risk of data leaks, and inventory overstock. Ideate & Prototype: Designed a streamlined single-page dashboard paired with automated low-stock warnings and strict role separation (Manager vs. Staff). Test & Iterate: Validated technical viability through extensive offline integration testing (db/tests/) to ensure concurrency handling and zero tenant-data spillover. The solution is fully deployed on Replit and Supabase, ready for Phase 2 real-world usage.

4. Preliminary Environmental & SDG Impact Framework

To align with SDC's focus on meaningful global sustainability, establish your project's impact baseline before building the prototype. Use estimated data, published research, or public benchmarks where exact figures are unavailable. Provide the citation of any sources used.

*The target, baseline, and KPI you provide here will be used by judges during shortlisting to assess **Innovation and Creativity** as well as **Feasibility of build**. Vague or unsupported projections may weaken your case for advancing to Phase 2.*

Metric Category	**Your Response**
Target SDG Sub-Target(s) *Identify the exact UN SDG sub-target your project addresses* *e.g., Target 11.6 "Reduce the adverse per-capita environmental impact of cities, focusing on air quality and waste management."*	Target 12.5: "By 2030, substantially reduce waste generation through prevention, reduction, recycling and reuse." (Also aligned with Target 9.4 for upgrading technological capabilities of small industries)
Current Problem Baseline *What is the starting condition of the problem, expressed as a number?* *e.g., "The target infrastructure currently experiences 35% energy inefficiency due to unoptimized scheduling."*	SMB inventory management currently suffers from 25% average material overstock/ waste and a 35% inefficiency rate due to unoptimized reordering and manual tracking errors.
Core Measurable KPI *The specific unit you will track* *e.g., % reduction in power consumption, processing-latency improvement, or kg of resource waste avoided.*	% reduction in inventory material waste and % reduction in stockout/overstock occurrence.
Phase 2 Impact Target *Your realistic, percentage-based target by the final prototype presentation* *e.g., "20% improvement in computational resource use."*	Achieve a 20% reduction in unnecessary inventory stocking waste and a 30% improvement in tracking accuracy/system latency.

5. Eligibility & Readiness Checklist

Tick all boxes that apply before submitting.

☑ All team members have completed the mandatory GenAI, Database, and Cloud Deployment upskilling workshops.

☑ The project concept addresses a verified real-world problem mapped directly to one or more UN SDGs.

☑ The proposed solution targets a functional, deployed application rather than a theoretical business case study.

☑ All fields in this template have been completed within the word limits stated.

6. IP & Originality Declaration

6.1 Pre-Existing / Prior-Owned Work

Does this submission include any code, design, research, or other work that any team member created or owned before the start of this hackathon?

☐ Yes ☑ No

If yes, briefly describe what was pre-existing and which team member(s) owned it:

6.2 Third-Party Code, Datasets, or Models

Does your project use any third-party code libraries, open-source frameworks, public/private datasets, or pre-trained AI models that your team did not create?

☐ Yes ☑ No

If yes, list them below (open-source use is permitted provided it is declared and properly licensed/attributed):

Name	Type (code/dataset/model)	License

6.3 Employment-Related IP Conflicts

Is any team member currently bound by an employment, internship, or consulting agreement under which their employer could claim ownership of work they create during this hackathon?

☐ Yes ☑ No

If yes, please describe the nature of the agreement and which member it concerns:

7. Declaration

By submitting this proposal, the team leader confirms that the information provided is accurate to the best of the team's knowledge and that the project will be original work created during the SDC 2026 hackathon period.

[Signature] 

[Date] 25/10/2026